Hrishikesh Pawar

Worcester, MA | +1(774)-519-2900 | hpawar@wpi.edu | linkedin.com/in/hp39 | hrishikesh-st.github.io

Education

Worcester Polytechnic Institute — Worcester, MA
Master of Science in Robotics Engineering; GPA: 4.0 — *August 2023 - May 2025*
Selected Coursework: Deep Learning, Computer Vision, Vision-based Robot Manipulation

Savitribai Phule Pune University — Pune, India
Bachelor of Engineering in Mechanical Engineering; GPA: 3.55 — *August 2016 - April 2020*

Skills

- **Languages**: Python, C, C++, SQL
- **Frameworks**: Pytorch, OpenCV, TensorFlow, ONNX, Numpy, Pandas, Django, Flask, Celery, Kafka, Matlab, CMake, Docker, TensorRT, Jenkins, Kubernetes, Git, ROS, ROS2

Experience

- **Nobi** | *AI Software Intern* | *Remote* — *May 2024 - August 2024*
  Vision Stack Development, Model Optimization and Automation Pipelines
  - Worked on real-time activity recognition vision stack using single fisheye images for indoor environments.
  - Investigated **LLaVA**, **BCLIP**, and **CLIP** for multi-modal vision-language tasks, using **LoRA** for **PEFT** fine-tuning.
  - Analyzed and deployed SOTA **Vision Transformers**, enhancing object detection performance from **85 mAP** to **92 mAP**.
  - Implemented **CI/CD** workflows using **Jenkins** and **Kubernetes**, automating model lifecycle from data pre-processing to training, evaluation and deployment in dockerized environments.
- **Adagrad AI** | *Computer Vision Engineer* | *Pune, India* — *November 2020 - July 2023*
  Smart Lamps
  - Led the R&D initiatives for vision platform for smart ceiling lamps, enabling **real-time fall detection** for elderly care.
  - Designed a scene understanding system using **person detection, action recognition** and **pose estimation**.
  - Proposed and implemented a novel approach using single fisheye camera implementing a **rotation-aware detection model**.
  - Achieved a **4x** reduction in memory usage, facilitating transition to low-compute edge-devices reducing **40%** production cost.
  - Developed training, monitoring and deployment pipelines using **Nvidia DALI**, **Triton**, **Docker**, and **AWS EC2**.

  Gate-Guard | Indian Express | Times of India | YouTube
  - Developed an edge-based Boom Barrier Automation system using **ALPR (Automatic License Plate Recognition)**.
  - Built data collection, training and deployment pipelines for lightweight object detection networks: **YoloV4, YoloX**.
  - Implemented data-augmentation techniques to enable color-invariant plate recognition, resulting in **20% mAP** improvement.

  Edge-Based Retail Management Application
  - Built an end-to-end edge web applications enabling customer visibility, inventory management, and staff optimization.
  - Developed data-collection pipelines employing **OpenCV, Flask** and **PostgreSQL** to collect **40,000** images per day.
  - Administered Docker containerization to ensure data security and portability across various **NVIDIA's** edge devices.

Projects

- ***Classical Structure from Motion Pipeline*** | GitHub
  - Implemented a Classical Structure from Motion (SfM) pipeline to reconstruct 3D scenes from 2D images.
  - Developed modules for Fundamental Matrix estimation, Pose estimation, Triangulation, and Bundle Adjustment.
  - **Tech Stack**: OpenCV, Numpy, SciPy
- ***Zero-Shot Semantic Neural Style Transfer for Images*** | GitHub
  - Developed a Neural Style Transfer system for images, inspired by the paper AdaAttN.
  - Integrated **AdaAttN** with **CLIPSeg** for zero-shot style transfer, enabling prompt-based, image segmentation.
  - **Tech Stack**: PyTorch, CLIP, custom training and inference scripts.
- ***Multi-Object Grasp Point Detection from Object Top-Surface using RGB-D images*** | GitHub
  - Designed a grasp point detection system for a parallel jaw gripper using depth images of the object top surfaces.
  - Developed an object geometry specific heuristic algorithm, leveraging 3D Delaunay triangulation for detecting stable grasps.
  - **Tech Stack**: ROS2 Humble, Gazebo, Point Cloud Library, Open3D